

WASP-43b

R-1049

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_160418 · created 2026-07-30T09:32:33+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457496.6641 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.057
Transit depth (Rp/Rs)^2	3.88 +/- 2.26 [%]
Orbital Inclination (inc)	82.92 +/- 2.74
Airmass coefficient 1 (a1)	1.2 +/- 0.22
Airmass coefficient 2 (a2)	-0.13 +/- 0.16
Scatter in the residuals of the lightcurve fit is	18.55 %
Best Comparison Star	#1 - [334, 160]
Optimal Method	PSF photometry
Transit Duration (day)	0.0534 +/- 0.0095

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.197 ± 0.057 vs 0.1594 (deviation 0.63σ)
Duration fit vs published	0.0534 d vs 0.0512 d (deviation 0.21σ)
Mid-time O–C	2.3 min
Depth SNR	3.3
Residual scatter	18.55 %

Light curve

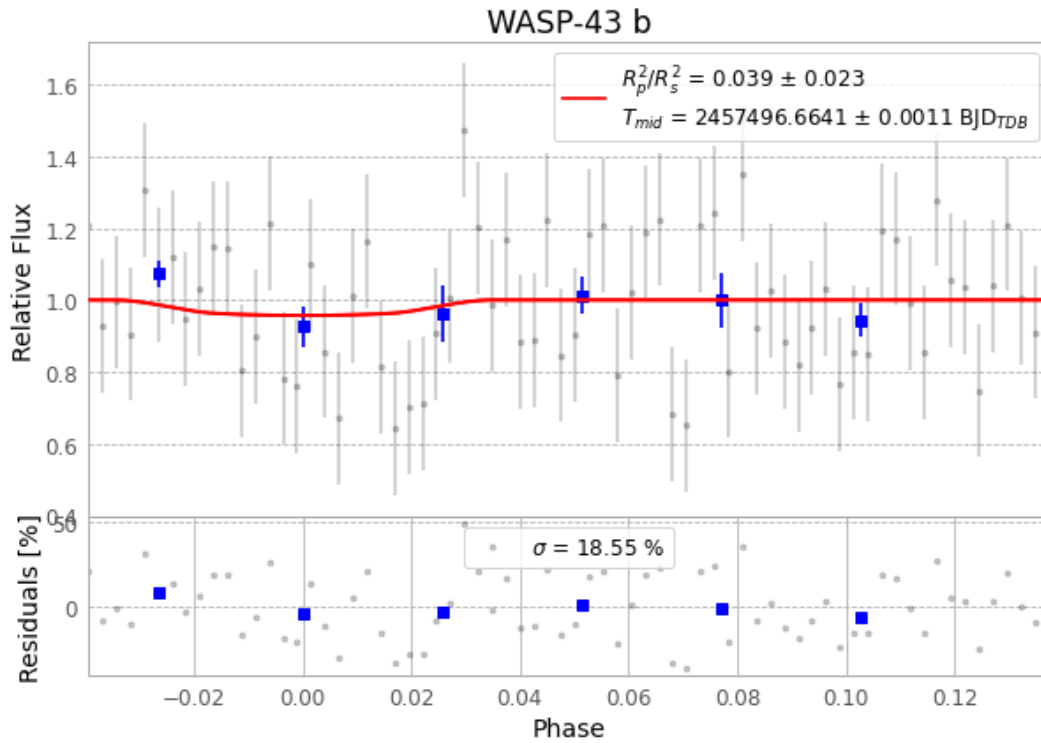


Figure 1 — FinalLightCurve_WASP-43 b_2016-04-17.png (SHA-256 9a5c52578815a3b0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [355, 171], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fdedb29a4bdf9186...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), WASP-43160418030312.FITS → WASP-43160418063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-160418.log (19be0f9faebf...), FinalLightCurve_WASP-43 b_2016-04-17.png (9a5c52578815...), FinalLightCurve_WASP-43 b_2016-04-17.csv (de6556b22424...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)